Neural Correlates of Memorability

Abstract

Memorability is a perceptual property of images that predicts how well they are remembered, driven by contents rather than low-level visual features. Using fMRI data from 22 participants performing a visual memory task, we explored neural correlates of memorability during encoding and recall phases via multivariate pattern analysis (MVPA) and representational similarity analysis (RSA). Results showed significant decoding of memorability levels across regions of interest (ROIs) during encoding, including object- and scene-selective areas, but reduced accuracy during recall, with medial temporal lobe regions showing no significance. RSA revealed strong correlations between brain-based and memorability-based models during encoding, which decreased during recall. These results suggest a distributed coding of memorability-related information during encoding but diminished resolution during recall.

Introduction

Memorability is an intrinsic, perceptual stimulus property that tells how well a given image will be later remembered or forgotten, independent of individual differences in memory performance or experimental context (Bainbridge, 2019). Studies have also demonstrated the memorability as a stable and consistent characteristic of an image, driven by their content rather than low-level visual features (Bainbridge et al., 2017). Functional magnetic resonance imaging (fMRI) studies have revealed distinct neural basis for memorability through contrasting memorable versus forgettable images. They found distributed activations from higher-order visual areas to memory related regions, but no significant difference in early visual cortex (Bainbridge, 2019). These findings further support the idea that memorability effect is not driven by low-level visual properties.

Here, we stretched the scope from perception/encoding to recall during visual imagery task, asking whether we can decode memorability-related information via multivariate pattern analysis (MVPA) and representational similarity analysis (RSA) across regions of interest (ROI). We first investigated the decodability of the memorability level (e.g., high versus low) for a given image category across ROIs during encoding and recall periods. Second, we looked for representational similarities between memorability-based and brain-based models for every ROI during encoding and recall.

Methods

Dataset. This dataset is from Bainbridge et al., where 22 participants (15 female; mead: 24 years, standard deviation: 3.4 years) performed a visual memory task in 7T MRI scanner (Bainbridge et al., 2020). Task consists of three phases: encoding (6 seconds), distractor (4 seconds), and recall (6 seconds visual imagery cue, 2 seconds response period), and vividness ratings for their mental imagery were reported as their behavioral measures. 192 images with nested categorical structure with different levels of information (e.g., coarse, mid, and fine) were used as main stimuli. For the preprocessing of the data, a GLM was run to model each of the fine-level categories on odd and even runs of the task separately. No memory trials were excluded from the final dataset, as well as correcting for motion artifacts.

Memorability Scores. Each fine-level image category consisted of 8 exemplar images. We used a dedicated neural network, ResMem, to get hypothetical memorability scores for each image within the category (Needell & Bainbridge, 2022). Then, we calculated the average memorability score for each fine-level image categories.

Analysis Technique 1. We used MVPA to access the decodability of memorability information across ROIs. For every subject, ROI, and phase, we got corresponding odd and even trial data from fMRI scan. Any trials that had missing brain data were excluded. Odd and even trial data were then concatenated to increase the datapoint for later classification, resulting in 2D matrix (trials x available ROIs) of normalized beta weights for a given condition. For the classification label, we calculated the hypothetical memorability scores for every image via ResMem and calculated the average scores for each fine-level category. To create a binary label for classification, we did a median split on the scores and marked 1 for high and 0 for low memorability.

With brain data and labels, we ran 10-fold Stratified Cross Validation using logistic regression and then averaged the accuracy scores from each of the split. To get the null distribution of classification accuracy, we shuffled the labels and ran same cross validation steps 100 times and got an average score as well.

For the statistical testing, we used 1 sample t-test against 0.5 for every ROI and corrected for multiple comparisons via Benjamini-Hochberg method. The choice of population mean as 0.5 was based on the average classification scores for the shuffled labels.

Analysis Technique 2. We used RSA to assess the representational similarities between representations in ROIs and memorability scores. To calculate the representational dissimilarity matrix (RDM) for memorability, we calculated the absolute difference in memorability scores between pair of fine-level image categories. To calculate brain-based RDMs, for every subject, ROI, and phase, we calculated the Pearson correlations between the voxel values from odd trial data and even trial data and vice versa. Then, we averaged the correlation values and subtracted from 1 to get the similarity for a given pair of fine-level image categories. For later comparisons between RDMs, we performed a fisher's z transform on the correlation values in RDMs.

To access the similarities between brain-based RDMs and memorability-based RDM, we first flattened the lower-half of RDM values into a vector. Then, we calculated a Spearman's correlation between RDM for a given ROI and RDM for memorability. Since we are only interested in the magnitude of similarity, we reported an absolute value of the resulting correlation values. For the null distribution, similar approach used in Analysis 1 was used, where we shuffled the vector output of brain-based RDM and calculated the Spearman's correlation for 100 runs.

For the statistical testing, we used 1 sample t-test against 0.05 for every ROI and corrected for multiple comparisons via Benjamini-Hochberg method. The choice of population mean as 0.05 was based on the average correlation values for the shuffled labels.

Results

First, we looked at individual memorability scores for fine-level images to see whether we see reasonably ranged spectrum of scores. Memorability scores ranged from 0.414 to 0.980 (mean = 0.748, standard deviation = 0.156). After performing median split on averaged memorability scores for fine-level image categories, 11 out of 12 fine-level image categories from objects category and 1 out of 12 from scene category were assigned to high memorability groups (Figure 1).

For our first analysis, we examined whether specific brain regions coded for memorability-related information (e.g., high or low memorability scores). The following ROIs were used in the analyses: lateral occipital (LO), posterior fusiform (PFS), parahippocampal place area (PPA), retrosplenial cortex (RSC), occipital place area (OPA), the head and body of the hippocampus (headbody), the tail of the hippocampus (tail), perirhinal cortex (PRC), and early visual cortex (EVC). During encoding phase, all ROIs showed significant accuracy results for decoding whether presented fine-level image category had high or low memorability scores (Figure 2A). However, during recall phase, classification accuracy decreased in overall, and hippocampal regions showed no significance, as well as PRC (Figure 2B; see table 1 & 2 for detailed statistics).

For our second analysis, we investigated the representational similarities between memorability-based model and brain-based data across ROIs (Figure 3) by looking at the Spearman's ranked correlation between two given RDMs. All ROIs, except for the headbody, showed significant correlation during encoding phase (Figure 4A). For the recall phase, similar to previous classification results, we saw an overall decrease in correlation values across all ROIs (Figure 4B), where only 4 ROIs showed significance (see Table 3 & 4 for detailed statistics).

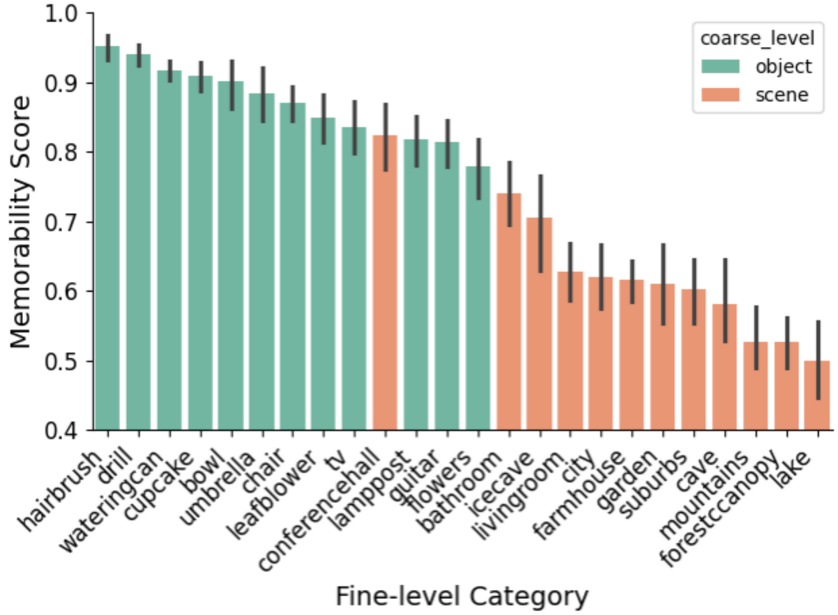
Discussion

The present study investigated the neural correlates of memorability in two ways. First, we conducted MVPA to classify the memorability level (e.g., high versus low) of a given fine-level image category during encoding and recall periods. We found significant classification accuracies across all ROIs during encoding, demonstrating a distributed coding of memorability-related information. We found significant but decreased classification accuracies across all ROIs, except for medial temporal lobe regions (headbody, tail, PRC) that showed no significance. This result aligns with the findings from the original paper, where all levels of information (fine, mid, and coarse) were not present in the hippocampus and slight discriminability found in PRC (Bainbridge et al., 2020). Since median-split memorability score distribution revealed similar coarse-level distinction (e.g., object versus scenes) between fine-level image categories, we would not expect medial temporal lobe regions to successfully discriminate between high versus low memorability image categories. However, it is surprising to see above chance accuracies in other regions, given that no visual stimuli were presented during recall period. Second, we investigated representational structures of ROIs by comparing them with memorability-based model. RSA again revealed similar results to first analysis, where we found significant correlations between brain-based and memorability-based RDMs across ROIs during encoding and decreased correlations overall during recall. Object selective regions (e.g., LO, PFS) and scene selective regions (e.g., PPA, RSC) showed significant correlations for both encoding and recall period. This result is in-line with the findings in the original paper, as these regions showed significant information discriminability (Bainbridge et al., 2020). Also, our memorability-based RDM demonstrated strong coarse-level discriminability (i.e., memorability scores for object category images were similar to each other, and vice versa for scene category images), so high correlation results from object and scene selective regions were expected.

However, several limitations existed in current study. First, memorability was not controlled properly across all information levels. Images from object category in general showed higher memorability scores in general compared to those from scene category. This confounds our results that memorability information is decodable from fMRI data. All the results could have been due to a robust object/scene information discriminability of ROIs as demonstrated in original paper. Second, the data only contained fine-level category data, so we couldn't analyze the memorability effect in finer resolution. Also, this led to fewer datapoints that we can utilize for classification.

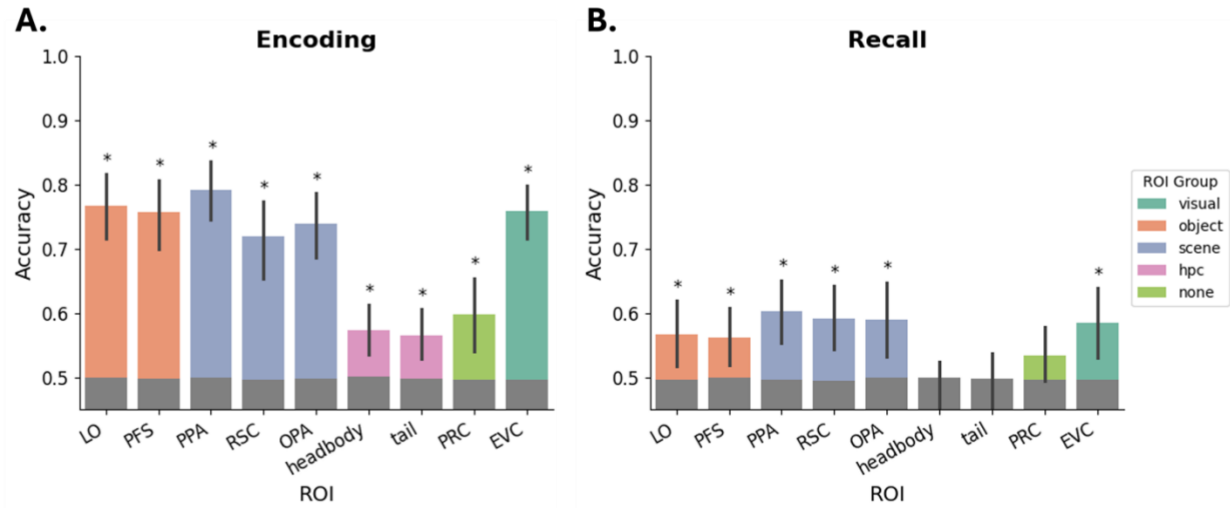
For future directions, we could investigate the relationship between memorability and vividness, whether we see higher vividness ratings for images that have higher memorability scores. We could also look at differences in decodability in aphantasics. It would be interesting to see whether we could still decode image-related neural activities given the participant's inability to perform visual imagery.

Figure 1. Memorability Scores for Fine-level Image Categories



Average memorability scores for fine-level image categories sorted by descending order. Error bars indicate 95% confidence intervals, calculated using 1000 iterations of bootstrapped resampling of the data.

Figure 2. Classification for memorability level across ROIs.



Classification accuracy for each ROI was calculated as average score across 10-fold Stratified cross validation for **A.** encoding phase and **B.** recall phase. Bar graphs indicate the mean accuracy for classifying memorability levels of fine-level categories. Significance (*) indicates results from a one-tailed t-test versus 0.5, with a FDR-corrected level of $q < 0.05$. Error bars indicate 95% confidence intervals, calculated using 1000 iterations of bootstrapped resampling of the data.

Figure 3. RDMs

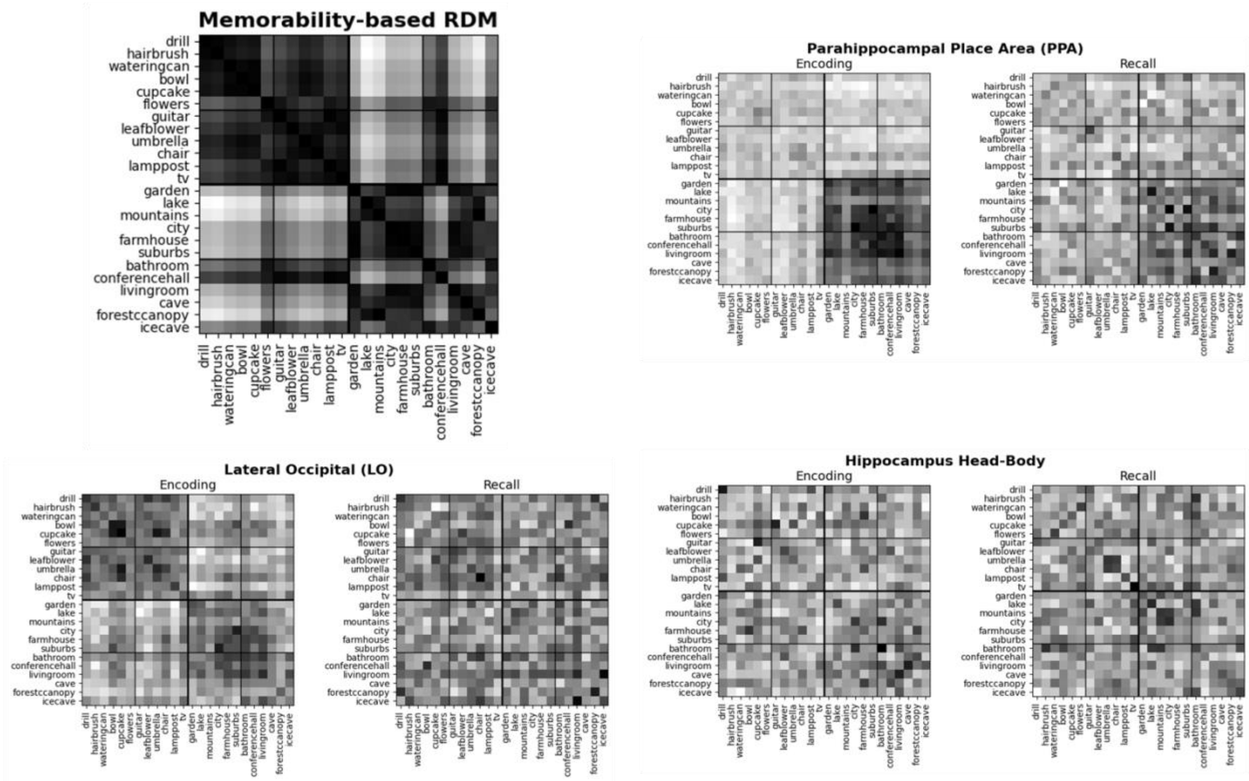
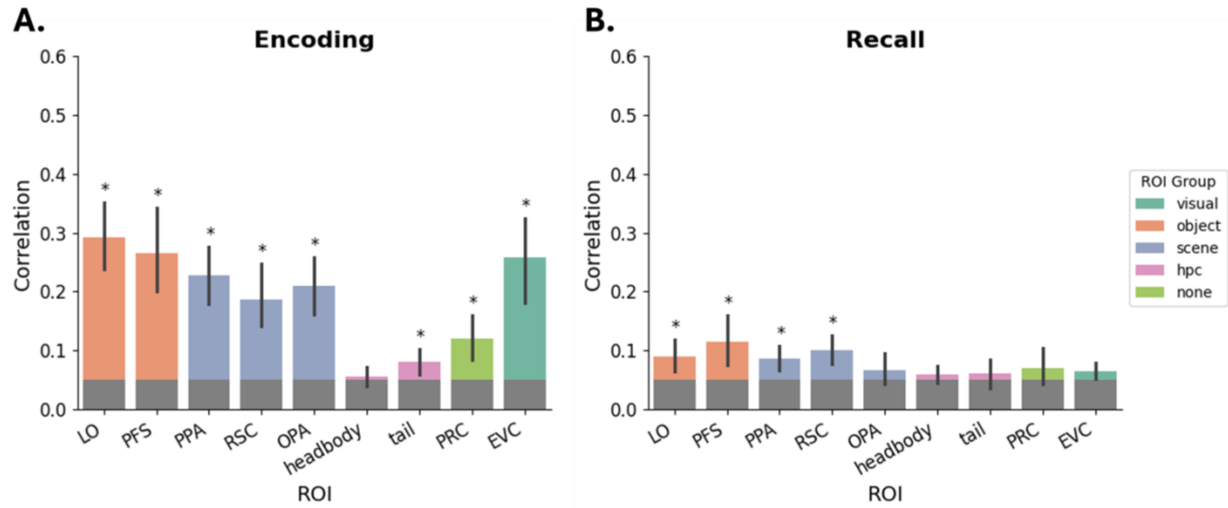


Figure 4. Correlations between brain-based and memorability-based RDMs across ROIs.



Spearman's rank correlation for each ROI was calculated for **A.** encoding phase and **B.** recall phase separately. Bar graphs indicate the mean correlation value between ROI-based RDM and memorability-based RDM. Significance (*) indicates results from a one-tailed t-test versus 0.5, with a FDR-corrected level of q < 0.05. Error bars indicate 95% confidence intervals, calculated using 1000 iterations of bootstrapped resampling of the data.

Table 1. Classification Performance for each ROI during Encoding

ROI	LO	PFS	PPA	RSC	OPA	Headbody	Tail	PRC	EVC
Accuracy	0.77	0.76	0.79	0.72	0.74	0.57	0.56	0.6	0.76
T-statistics (df=21)	10.82	9.55	12.25	7.1	8.88	3.62	3.21	3.42	11.81
P values	<0.001	<0.001	<0.001	<0.001	<0.001	<0.01	<0.01	<0.01	<0.001

Table 2. Classification Performance for each ROI during Recall

ROI	LO	PFS	PPA	RSC	OPA	Headbody	Tail	PRC	EVC
Accuracy	0.57	0.56	0.60	0.59	0.59	0.48	0.49	0.53	0.58
T-statistics (df=21)	2.45	2.72	4.10	3.39	3.05	-0.92	-0.33	1.5	2.91
P values	<0.05	<0.05	<0.001	<0.01	<0.01	0.817	0.706	0.097	<0.01

Table 3. Spearman's Rank Correlation for each ROI during Encoding

ROI	LO	PFS	PPA	RSC	OPA	Headbody	Tail	PRC	EVC
Accuracy	0.29	0.27	0.23	0.19	0.21	0.05	0.08	0.12	0.26
T-statistics (df=21)	8.23	5.76	6.89	5.07	6.17	0.59	2.67	3.42	5.56
P values	<0.001	<0.001	<0.001	<0.001	<0.001	0.281	<0.01	<0.01	<0.001

Table 4. Spearman's Rank Correlation for each ROI during Recall

ROI	LO	PFS	PPA	RSC	OPA	Headbody	Tail	PRC	EVC
Accuracy	0.09	0.11	0.09	0.1	0.07	0.06	0.06	0.07	0.06
T-statistics (df=21)	2.45	2.72	4.10	3.39	3.05	-0.92	-0.33	1.5	2.91
P values	<0.05	<0.05	<0.01	<0.01	0.155	0.155	0.207	0.156	0.056

References

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